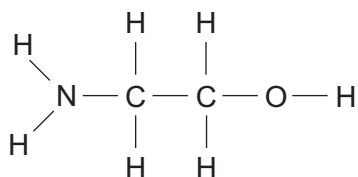


11. (a) 2-Aminoethanol is a colourless liquid.



2-aminoethanol

It is very soluble in water owing to its ability to hydrogen bond with water molecules.

Complete the diagram above to show how 2-aminoethanol is able to hydrogen bond to water molecules. [4]

- (b) The boiling temperatures of 1,2-diaminoethane and 2-aminoethanol are shown in the table.

Compound	Formula	Boiling temperature / °C
1,2-diaminoethane	$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$	117
2-aminoethanol	$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{OH}$	170

Use these figures to comment on the strength of the intermolecular forces between the molecules in each compound suggesting reasons for your answer. [3]

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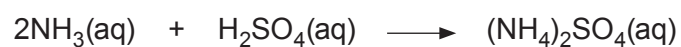
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- (c) Ammonium sulfate can be prepared in the laboratory by neutralising aqueous sulfuric acid with ammonia solution in an acid-base titration.



Outline a method to obtain pure dry crystals of ammonium sulfate by this procedure.

[6 QER]

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